Problem 5.11

Let I_e denote the interest earned using the exact simple interest method and I_b the interest earned using the Banker's Rule. Find the ratio $\frac{I_e}{I_b}$. Assume a non-leap year.

Using exact simple interest,

$$I_e = Pr\left(\frac{t}{365}\right).$$

Using Banker's Rule,

$$I_b = Pr\left(\frac{t}{360}\right).$$

Now compute the ratio:

$$\frac{I_e}{I_b} = \frac{Pr\left(\frac{t}{365}\right)}{Pr\left(\frac{t}{360}\right)}.$$

Canceling the common factors P, r, and t, we get

$$\frac{I_e}{I_b} = \frac{\frac{1}{365}}{\frac{1}{360}} = \frac{360}{365}.$$

Therefore,

$$\frac{I_e}{I_b} = \frac{360}{365} \approx 0.9863.$$